

JESS Thermodynamic Database v8.9

Berkelium

24-Jan-23

Reaction No. 2711							
F-1 + Bk+3 = Bk+3_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.788(4SF)	1	EOR	
2	25	1	NaClO ₄	lgK:2.89(3SF)	5	MDS	437
3	10:50	1	Unknown	dH:3.(1SF)	6	CRV	552
4	25	1	NaClO ₄	dH:6.0(5.SD)	0	EVK	437

Reaction No. 4732							
Bk+4 + e-1 = Bk+3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.67(3SF)	4	CRV	6330
2	25	0	Inf. Dilution	E:1.67(3SF)	4	CRV	22519
3	25	0	Inf. Dilution	E:1.67(3SF)	4	CRV	22551
4	25	1	HClO ₄	E:1.6(0.1SD)	5	ENS	775

Reaction No. 4733							
Bk+3 + 3<e-1> = Bk(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:139.(2.SD)	2	ENS	3006
2	25	0	Inf. Dilution	dG:574.(3SF)kJ	2	CRV	7421
3	25	0	Inf. Dilution	E:-2.4(0.1SD)	0	MDG	775
4	25	0	Inf. Dilution	E:-1.97(3SF)	0	ENS	7276
5	25	0	Inf. Dilution	E:-2.0(2SF)	2	CRV	22551
6	25	0	Inf. Dilution	dH:144.(1.SD)	2	ENS	3006

Reaction No. 6334							
Oxalic-2 + Bk+3 = Bk+3_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.738(4SF)	1	EOR	
2	25	0.1	Unknown	lgK:5.5(0.1SD)	4	MTP	6545

Reaction No. 6335							
Bk+3_Oxalic-2 + Oxalic-2 = Bk+3_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:3.7(0.05SD)	4	MTP	6545

Reaction No. 6611							
Acetic-1 + Bk+3 = Bk+3_Acetic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0.70001	2	NaClO4	lgK:1.77(3SF)	6	MDS	2112
2	10.4	2	NaClO4	lgK:1.98(3SF)	4	MDS	2112
3	25	0	Inf. Dilution	lgK:2.802(4SF)	1	EOR	
4	25	2	NaClO4	lgK:2.05(3SF)	6	MDS	2112
5	25	2	Unknown	lgK:2.06(3SF)	3	EOD	31112
6	39.8	2	NaClO4	lgK:2.18(3SF)	6	MDS	2112
7	25	2	NaClO4	dG:-2.79(0.01SD)	5	ENS	2112
8	25	2	NaClO4	dH:4.4(0.1SD)	5	MCL	2112
9	25	2	NaClO4	dS:24.(0.3SD)	5	EFE	2112

Reaction No. 12160							
Citric-3 + Bk+3 = Bk+3_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.772(4SF)	1	EOR	
2	25	0.1	Unknown	lgK:7.9(0.08SD)	4	MDG	6545

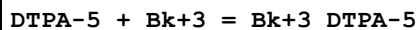
Reaction No. 12161							
Bk+3_Citric-3 + Citric-3 = Bk+3_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:3.3(0.2SD)	3	MDG	6545

Reaction No. 12162							
Bk+3_Citric-3 + H+1_Citric-3 = Bk+3_H+1_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:2.6(0.1SD)	4	MDG	6545

Reaction No. 14647							
CDTA-4 + Bk+3 = Bk+3_CDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

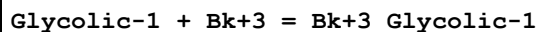
1	25	0	Inf. Dilution	lgK:21.6(3SF)	1	EDH	
2	25	0.1	NH4ClO4	lgK:19.16(4SF)	2	MIX	999
3	25	0.1	Unknown	lgK:20.1(0.03SD)	2	CRV	552

Reaction No. 14860



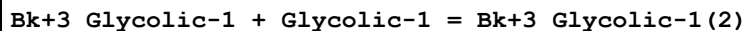
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH4ClO4	lgK:22.79(4SF)	6	MIX	999

Reaction No. 24480



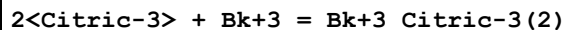
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0.5	2	NaClO4	lgK:2.73(3SF)	6	MDS	2110
2	10.1	2	NaClO4	lgK:2.70(3SF)	6	MDS	2110
3	25	2	NaClO4	lgK:2.65(3SF)	6	MDS	2110
4	40.6	2	NaClO4	lgK:2.62(3SF)	6	MDS	2110
5	52.8	2	NaClO4	lgK:2.57(3SF)	6	MDS	2110
6	25	2	NaClO4	dH:-1.21(3SF)	5	MCL	2110

Reaction No. 24481



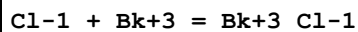
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0.5	2	NaClO4	lgK:1.91(3SF)	6	MDS	2110
2	10.1	2	NaClO4	lgK:1.94(3SF)	6	MDS	2110
3	25	2	NaClO4	lgK:2.05(3SF)	6	MDS	2110
4	40.6	2	NaClO4	lgK:1.90(3SF)	6	MDS	2110
5	52.8	2	NaClO4	lgK:1.98(3SF)	6	MDS	2110

Reaction No. 45982



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.07(4SF)	1	EOR	
2	25	0.1	Unknown	lgK:11.2(3SF)	4	CRV	2556

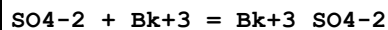
Reaction No. 53017



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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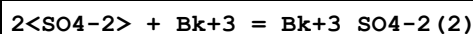
1	20	3	LiClO4	lgK:-0.23 (2SF)	5	CRV	2556
2	25	0	Inf. Dilution	lgK:1.196 (4SF)	1	EOR	
3	25	1	NaClO4	lgK:-0.02 (1SF)	4	MDS	5398

Reaction No. 53323



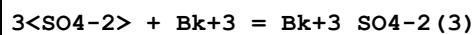
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.71 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:3.71 (3SF)	4	MDS	5398

Reaction No. 53324



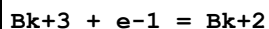
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.59 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:5.59 (3SF)	4	MDS	5398

Reaction No. 53325



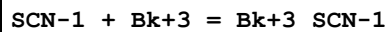
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.04 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:5.04 (3SF)	4	MDS	5398

Reaction No. 55643



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.8 (2SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:-2.8 (2SF)	4	ENS	7276
3	25	0	Inf. Dilution	E:-2.8 (2SF)	4	CRV	22519
4	25	0	Unknown	E:-2.5 (0.2SD)	0	ENS	5256

Reaction No. 57384



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	5	ClO4-1 salt	lgK:0.9 (0.04SD)	5	MSE	5243
2	25	0	Inf. Dilution	lgK:1.534 (4SF)	1	EOR	
3	25	1	NaClO4	lgK:0.5 (0.03SD)	5	MSE	5612
4	25	1	NaClO4	lgK:0.49 (2SF)	5	MDS	11516

Note (4): Harmon H et al., J. Inorg. Nucl. Chem., 1972, 34, 1381							
5	25	5	ClO4-1 salt	lgK:0.9(0.04SD)	5	MSE	5243
6	40	5	ClO4-1 salt	lgK:0.9(0.04SD)	5	MSE	5243
7	55	5	ClO4-1 salt	lgK:0.9(0.05SD)	5	MSE	5243
8	10:60	5	Unknown	dH:1.(1SF)	6	CRV	552

Reaction No. 57387							
$\text{Bk(s)} + 0.5\langle\text{N}_2(\text{g})\rangle + \text{C(s)} + \text{S(s)} = \text{Bk+3_SCN-1} + 2\langle\text{e-1}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1.2(0.04SD)	0	ENS	5243
2	25	0	Inf. Dilution	dH:0.68(0.4SD)	0	ENS	5243

Reaction No. 60188							
$\text{EDTA-4} + \text{Bk+3} = \text{Bk+3_EDTA-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.32(4SF)	1	EOR	
2	25	0.1	NH4ClO4	lgK:18.9(0.2SD)	4	MIX	903
3	25	0.1	Unknown	lgK:18.5(3SF)	6	CRV	552

Reaction No. 60427							
$\text{OH-1} + \text{Bk+3} = \text{Bk+3_OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.224(4SF)	1	EOR	
2	25	0.1	Unknown	lgK:6.6(2SF)	4	CRV	2556

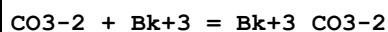
Reaction No. 60710							
$\text{Br-1} + \text{Bk+3} = \text{Bk+3_Br-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	3	LiClO4	lgK:-0.8(1SF)	4	CRV	2556

Reaction No. 66197							
$\text{TriMDTA-4} + \text{Bk+3} = \text{Bk+3_TriMDTA-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:14.36(4SF)	6	CRV	552

Reaction No. 70274							
$\text{Bk+2} + 2\langle\text{e-1}\rangle = \text{Bk(s)}$							

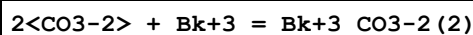
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.6(2SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:-1.6(2SF)	4	ENS	7276
3	25	0	Inf. Dilution	E:-1.6(2SF)	4	CRV	22519

Reaction No. 72331



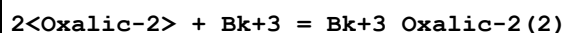
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.765(4SF)	1	EOR	
2	25	0.7	NaClO4	lgK:5.95(3SF)	4	EFE	8782

Reaction No. 72332



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.62(4SF)	1	EOR	
2	25	0.7	NaClO4	lgK:10.2(3SF)	4	EFE	8782

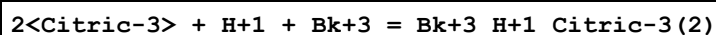
Reaction No. 75933



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.85(4SF)	1	EOR	
2	25	0.1	Unknown	lgK:9.19(3SF)	5	MTP	11516

Note (2): Stepanov A, Zhur. Neorg. Khim., 1971, 16, 2981

Reaction No. 77909



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.71(4SF)	1	ELC	

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability

5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EDH	Debye-Hueckel corrections
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
EVK	Variation of lgK with T
MCL	Calorimetry
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MIX	Ion exchange
MSE	Solvent Extraction
MTP	Migration or transference number

References

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